Flight Dynamics of the Sikorsky UH-60A Black Hawk in JSBSim

An individual-blade rotor model after Sikorsky's GenHel, checked against US Army flight tests

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Abstract

A flight-dynamics model of the UH-60A Black Hawk for the FlightGear simulator has been written in JSBSim, replacing the parametric YASim helicopter model the aircraft used before. It follows the structure of the Sikorsky General Helicopter (GenHel) engineering simulation documented in NASA CR-166309: the four main rotor blades are simulated individually, each with flap and lead-lag freedom, a nonlinear lag damper and eight aerodynamic segments using the SC1095 tables over ±180° and Mach number, under a Peters–He finite-state dynamic wake; the tail rotor is GenHel's closed-form solution; fuselage, stabilator and fin use GenHel's wind-tunnel tables with the rotor wake interference; the automatic flight control system (stabilator schedule, SAS, pitch bias actuator, flight path stabilization) and two T700 engines with the rotor speed degree of freedom complete it. The whole model, 797 aerodynamic functions and 380 control-system components, is generated by a script from transcribed data, and runs in plain JSBSim without scripting. It was flown offline through a scripted programme and compared with US Army (USAAEFA) flight tests and with GenHel. Level-flight trims from 50 to 162 kt agree with the test within 7.0 % of longitudinal and 7.0 % of collective travel and 1.9° of pitch attitude. Out-of-ground-effect hover power is 0.965–0.995 of the tethered-hover test, rotor figure of merit is within 0.026 of the measured data, and ground-effect power savings are within 0.010. For the 17 control-step responses of NASA TM-88360 the model's root-mean-square rate error is below GenHel's in 12 of 16 usable cases (mean 2.14 against 2.77 deg/s). Deficiencies are quantified: cruise power 2–10 % above one of two conflicting test sources, a weak pitch-up after an aft step at 100 kt (16.4 against 24.1 deg/s), an angle-of-attack derivative M_w a third of GenHel's, and half the bank change between climb and descent that the aircraft shows.

1 Introduction

The UH-60A distributed for FlightGear flies on YASim, whose helicopter model builds the rotor from a short list of geometric parameters and tuning factors. Its behaviour is set by tuning rather than by the aircraft's data, and its couplings, power characteristics and responses cannot be traced to measurements. For the UH-60A, unusually, the data exist in public: in the early 1980s Sikorsky documented its General Helicopter (GenHel) engineering simulation of the Black Hawk in full for NASA [1], NASA re-implemented and validated it against US Army flight tests [2, 3], and the Army's test reports [6, 5, 7] give trim, stability and performance data for the same airframes.

This report describes a JSBSim [18] model built on that material and how well it reproduces the tests. GenHel is used as the blueprint, not as gospel: where it leaves the physics incomplete (the induced power, Section 5.3) or where validation exposed an inconsistency, the model departs from it, and the departures are listed in Section 6. All of the flight model, including the rotor's integration, the control laws and the engines, is JSBSim XML; the FlightGear side only animates and instruments it. All result tables, and the numbers quoted from them, are written by the evaluation scripts.

2 Reference data

Table 1 lists the sources; most are scans, and every curve used was digitised from them. A few points of data reduction matter for the comparisons. *(i)* The control-step responses of TM-88360 were digitised from 68 plot pages (some scanned sideways or upside down); each case is replayed at its own test condition from table C1 of that report: weight, centre of gravity (CG), density altitude and speed. 8 of the cases, all five in hover among them, were flown at FS 359.2–359.6, near the aft limit; for the aft step at 100 kt this raises the pitch response by half. *(ii)* The trims of TM-88360 fig. B1 come from USAAEFA test 79-24; its own figure 1 shows that the three points plotted at 22–32 kt were flown at 3–15 KCAS, so they are near-hovers and are shown apart. *(iii)* Two tests of the same airframe, S/N 77-22716, disagree on cruise power: at 95 kt the main rotor torque of 79-24 (TM-88360) implies 4–6 % more power than the engine torque of 77-17, of which instrumentation drag explains about 2 %. *(iv)* The hover charts of the operator's manual represent the sixth-year aircraft with the external stores support system (ESSS) fairings, which needs about 3 % more hover power than the first-year aircraft; the model is compared with the latter.

3 The model

3.1 Implementation

A Python script writes the model from the transcribed tables (Table 2). JSBSim evaluates aerodynamic functions on demand, so the function graph must be free of cycles; every quantity that depends on its own past, the blade flap and lag angles and rates, the wake states, the rotor speed, the tail rotor inflow, the blade's elastic twist and the engines' states, therefore lives in flight-control-system channels, which JSBSim runs once per frame in a fixed order and which can write properties back. These channels integrate the states explicitly at the simulation rate of 480 Hz, which is 3.2° of rotor azimuth per step; the wake states use a semi-implicit step. Trims and control responses at this rate are within 0.5 % of those at 960 Hz, where 120 Hz was off by up to 4 %, and one step costs about 0.1 ms. The blade loads of a frame use the states of the previous frame. The same XML runs in FlightGear and in the offline harness used here.

3.2 Main rotor

Each blade is a rigid beam on a coincident flap and lag hinge at 4.66 % radius. Its eight segments of equal annulus area (Fig. 1b) see the air velocity at their mid-point from the full

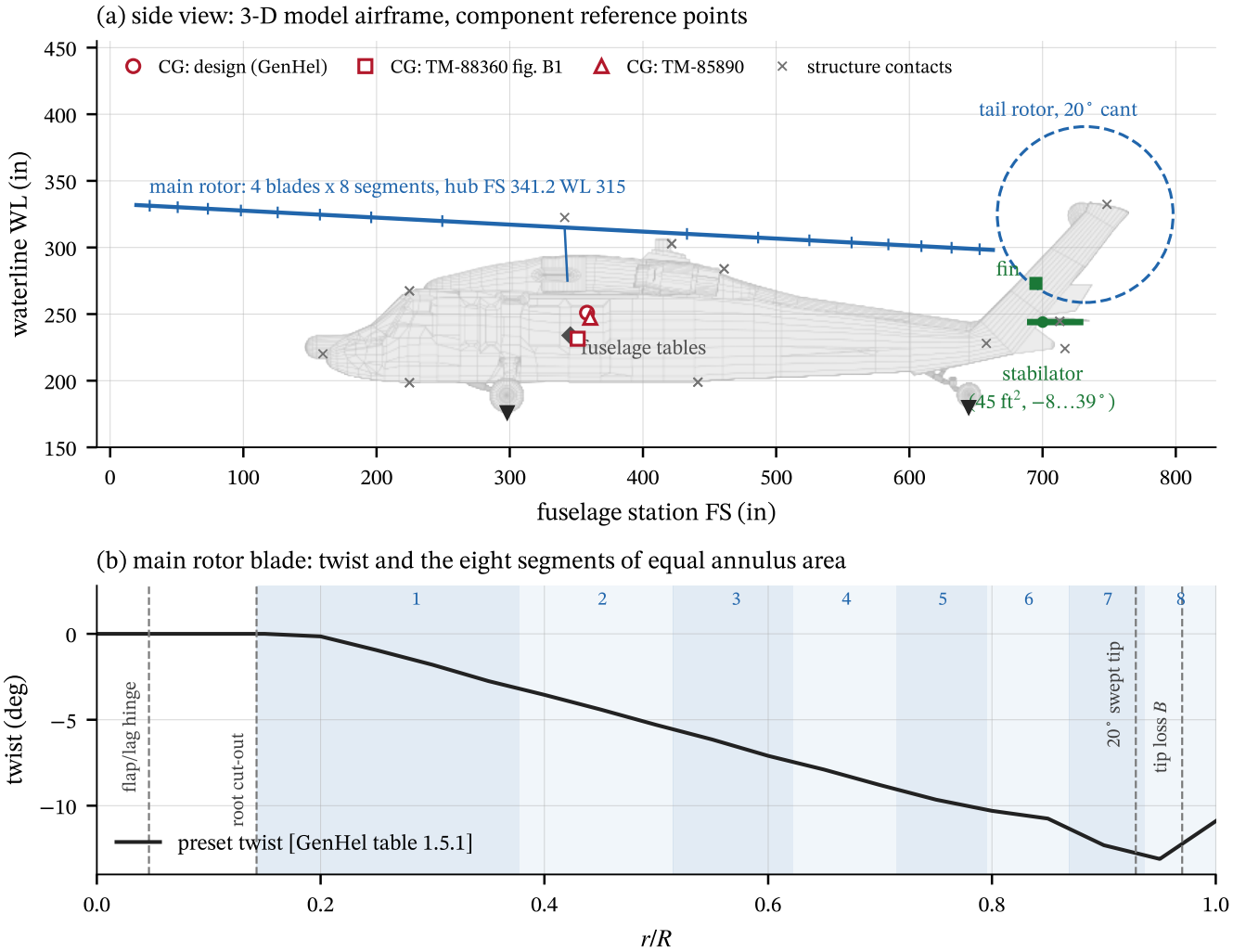


Figure 1: Geometry. (a) The FlightGear 3-D model's airframe in Sikorsky stations, with the main rotor disc (3° forward shaft tilt; ticks at the segment mid-points), the tail rotor, the stabilator and fin reference points, the reference point of the fuselage tables, the three centres of gravity used in this report, the wheels (▼) and the 17 structure contacts. (b) The main rotor blade: preset twist and the eight segments of equal annulus area.

kinematics in shaft axes: the aircraft's translation and rotation, the rotor speed, the blade's flap and lag motion and the induced velocity. Lift and drag come from the SC1095 tables of GenHel against angle of attack (±180°; the drag of the reversed section near 180°, which GenHel's table sets to zero, is restored) and Mach number, which on the 20° swept tip is that of the velocity normal to the leading edge. Lift acts on the velocity normal to the span; drag acts along the whole flow including its spanwise part, which is what gives the blade's profile power its growth with advance ratio. A section drag increment of 0.0012 is added to the smooth wind-tunnel tables: a production blade's boundary layer is turbulent from near the leading edge, and tripping added about half of the NACA 0012 increment [10] to the SC1095 at full-scale Reynolds number [9]. GenHel's empirical torsional wind-up twists the tip nose-down in proportion to the mean blade load. The tip loss factor $B = 0.97$ removes the lift outboard of BR and, unlike GenHel, also enters the momentum balance (Section 5.3).

The flap equation of each blade contains the aerodynamic hinge moment, the hub's specific force, the gyroscopic terms from the body's rate and acceleration, the centrifugal stiffness (including the lead rate) and the lag damper; the blade rests on droop and anti-flap stops. The lag equation contains the aerodynamic and damper moments, the centrifugal stiffness of the hinge offset, the Coriolis moment of flapping and the rotor's acceleration. The lag damper's force comes from GenHel's force–stroke-speed table and acts through the geometry of its two pickups, one on the hub and one on the blade cuff, which turns with blade pitch, so that its flap and lag moments follow from virtual work. The blades' aerodynamic and inertial loads act on the hub at their hinges; the shaft torque drives the rotor speed.

3.3 Inflow

The induced velocity comes from the Peters–He generalised dynamic wake [14] with three axisymmetric states, giving the radial distribution in hover, and the first harmonic pair, driven by the rotor's hub moments and coupled to the wake skew. Its inflow gain matrices are tabulated against the wake skew parameter. The states are kept in the hub frame and turned into the frame of the in-plane flow for their update, so that a wake that swings round with the flow direction in a hover does not carry the states with it. Ground effect scales the induced velocity: in hover by $(1+0.3(R/Z)^2)^{-1/2}$, with Z the height of the hub, which reproduces the power savings of the operator's manual hover chart [15] (GenHel's own form gives half of

Table 1: Sources and what they contribute.

Source	Content used
NASA CR-166309 [1]	GenHel: rotor, tail rotor, fuselage, empennage, controls, AFCS, engine equations; SC1095, fuselage and empennage tables; physical and mass data
NASA TM-85890 [2]	parameter tables, control rigging and mixing; GenHel stability derivatives at 100 kt
NASA TM-88360 [3]	level-flight trims vs. airspeed (fig. B1); 17 control-step responses (app. D) with test conditions (table C1)
NASA CR-177360 [4]	GenHel against Army tests: sideslip sweeps; tail-rotor wash on the fin
USAAEFA 79-24 [5]	the test behind TM-88360: inherent sideslip, lateral-directional sweeps, static and long-term stability
USAAEFA 77-17 [6]	tethered hover, tail rotor shaft power, level-flight power (first-year aircraft)
USAAEFA 83-24 [7]	hover fits of 77-17 reanalysed, sixth-year aircraft with ESSS fairings
DTIC ADA180404 [8]	sideslip in climbs and descents
NASA TP-2003-212265 [9]	SC1095 geometry, tip sweep, drag at full-scale Reynolds number, tripping increment
LRTA paper [11]	UH-60 rotor figure of merit from flight and wind tunnel tests
JAHS 2004 paper [12]	accessory power; flat-plate drag area of the UH-60A
NASA TN D-234 [13]	ground effect in forward flight
METU thesis [14]	Peters–He finite-state wake equations
TM 1-1520-237-10 [15]	hover chart (ground effect, power), limits, procedures, AFCS functions
NASA T700 study [16]	engine and fuel control dynamics

Table 2: The generated model: JSBSim system files, their content and the number of control-system components or named aerodynamic functions.

file	content	items
controls	rigging, mixing, servos, stabilator schedule, SAS, pitch bias actuator, FPS	122
rotor	blade flap and lag, lag dampers, Peters–He wake, tail rotor inflow, elastic twist, wake turbulence	128
drive	two T700 engines and their fuel controls, rotor speed, accessories, rotor brake	72
outputs	FlightGear outputs: rotor animation, instruments, vibration cue	58
aero	blade segments (4×8), tail rotor, fuselage, stabilator, fin, interference	797

them), and in forward flight by the ratio of forward-flight to hover interference of TN D-234 [13] against the angle of the wake. For use in the simulator, coloured noise can be added to the inflow and its gradients in the regimes where the real wake is unsteady (ground effect, translational lift, vortex ring); its intensities are estimates and it is off in all results here.

3.4 Tail rotor

The tail rotor uses GenHel's closed-form (Bailey) solution for a rotor with linear twist [1]: thrust and torque coefficients in terms of the inflow and advance ratios, with pitch–flap coupling ($\tan\delta_3 = 0.70$), a tip loss of 0.92 and a fin blockage factor of 0.796 below 30 kt. Its inflow equation is solved by a relaxed fixed-point step per frame, again with the momentum taken over the effective disc. The rotor is canted 20° so that it also lifts, and sees the main rotor's wake and the fuselage's sidewash.

3.5 Airframe

The fuselage forces and moments come from GenHel's tail-off wind-tunnel tables against angle of attack and sideslip, evaluated with the local flow that includes the main rotor wake through GenHel's interference factors (functions of the wake skew and the disc's tilt). In rearward flight the tables are mirrored front to back. The stabilator and fin are strips with their own lift and drag tables over ±180°, the rotor wake, the fuselage's downwash, sidewash and dynamic-pressure loss at the tail. GenHel's empirical downwash correction terms [1], a side force, a rolling moment and a sideslip-dependent pitching moment for the wake's swirl at the empennage, from rotor-on wind tunnel tests and flight test, complete the airframe.

3.6 Flight controls

The rigging and the mechanical mixing (collective to cyclic and pedal, pedal to longitudinal cyclic) follow TM-85890 [2]; the primary servos are second-order systems. The automatic flight control system is GenHel's: the stabilator scheduled on airspeed and collective with pitch rate and lateral acceleration feedback; the two stability augmentation channels (SAS 1 analogue, SAS 2 digital, 10 % authority together); the pitch bias actuator; and the flight path stabilization (FPS) with attitude, airspeed and heading hold and turn coordination through the trim actuators. FlightGear cannot move the pilot's stick, so the FPS moves the stick's trim point instead, and the pilot's trim release becomes automatic when a control moves away from it. Following TM-88360 and CR-177360, the tail rotor rigging carries 0.5 in of left pedal that GenHel's rigging lacks, the effect of the tail rotor's wash on the fin.

3.7 Engines, drive and mass

Each T700-GE-700 is modelled after GenHel's engine module and the NASA integration study [16]: the hydromechanical unit meters fuel within acceleration and deceleration schedules to hold a gas generator speed reference from the power lever or the collective's load-demand spindle, trimmed by the power-turbine speed governor; the gas generator follows with its speed-dependent lag and the power turbine's torque depends on both speeds. Freewheel clutches connect the engines to the main gearbox; the rotor speed is a degree of freedom with the inertia of hub, drive and blades. The gearbox absorbs 3 % of the main rotor torque (assumed) and the accessories 65.8 hp, Sikorsky's figure [12]. Table 3 lists the principal data. The default loading is GenHel's design point; each validation case uses its own test loading. The landing gear has two main wheels and a castering tail wheel; 17 struc-

Table 3: Principal data.

Main rotor [1]	
radius; blades; chord	26.83 ft; 4; 1.73 ft
solidity; Lock number (sea level)	0.0821; 8.1
speed (100 %); tip speed	27.0 rad/s; 724 ft/s
flap/lag hinge offset; aerodynamic root	1.25 ft (4.7 %); 3.83 ft
blade weight; flap inertia	256.9 lb; 1513 slug ft^2
airfoil; twist; tip	SC1095 (GenHel tables); −18° nonlinear; 20° sweep outboard of 0.929 R
segments	8 of equal annulus area, mid-points 0.29…0.97 R
tip loss B; section drag increment	0.97; 0.0012
shaft tilt; swashplate phase	3° forward; −9.7°
lag damper	nonlinear force–speed table [1], pickup kinematics
Tail rotor [1, 2]	
radius; blades; solidity	5.5 ft; 4; 0.1875
speed; cant; tan δ_3	124.62 rad/s; 20°; 0.70
twist; tip loss; fin blockage	−18°; 0.92; 0.796 below 30 kt
Airframe [1, 3]	
design gross weight; CG	16001 lb; FS 358, WL 251
inertias $I_{xx}, I_{yy}, I_{zz}, I_{xz}$	5462, 39469, 36887, 1574 slug ft^2
stabilator; fin	45 ft^2, −8 . . . +39° TED; 32.3 ft^2
Drive [1, 12]	
engines; transmission	2 × T700-GE-700, 1553 shp; 2828 shp
rotor inertia (hub, drive, blades)	7911 slug ft^2
accessories; gearbox loss	65.8 hp; 3 % of main rotor torque

ture contacts taken from the 3-D model's skin carry the aircraft when it lands on its belly or rolls over.

4 Method

The evaluation drives JSBSim from Python with the same files as FlightGear, at 480 Hz, with the wake noise off. A trim solves for the four controls and the attitudes that zero the six accelerations averaged over two rotor revolutions, with the airframe held while the rotor and wake settle. Atmospheres reproduce each test's pressure and density altitude (JSBSim's temperature offset and the geometric altitude are iterated). The programme:

- *Level-flight trims* from hover to 162 kt at the loading of TM-88360 fig. B1 (16 000 lb, FS 351, WL 231.5, 5250 ft);
- *lateral-directional trims*: zero-sideslip and wings-level trims at 60, 100 and 140 kt, in 1500 fpm climbs and descents and with the CG 2.1 in right and 4.1 in left, against the digitised sweeps of 79-24 and ADA180404;
- *power*: out-of-ground-effect hover from 11 000 to 22 000 lb at the 2302 ft test site; level flight at $C_T \times 10^4$ = 65 to 90 and μ = 0.10 to 0.32; the isolated rotor's figure of merit at sea level; and blade-element momentum calculations of the same blade as a reference;
- *ground effect*: the test autopilot (written for the FlightGear flight tests) hovers the model at 5, 10, 20, 40 and 100 ft wheel height;
- *control responses*: the recorded control increments of the 17 cases of TM-88360 app. D, applied open loop from a trim at the test condition with the SAS, FPS and pitch bias actuator off and the stabilator fixed, as in the test;
- *stability derivatives* at 100 kt at the loading of TM-85890 (16 400 lb, FS 360.4), by central differences of the aerodynamic forces with the airframe held.

Table 4: Level-flight trim: model minus test (TM-88360 fig. B1); power as model/test −1. *Flown at 3–15 KCAS.

kt	lon	lat	coll	ped	θ	ϕ	power
		% of travel			deg		%
22*	−1.7	−4.3	−7.2	−0.6	+1.6	−1.2	−12
30*	−2.3	−12.5	−11.6	+4.8	+0.5	+1.3	−23
32*	−3.0	−12.2	−14.4	+7.5	+0.8	−0.6	−27
50	+3.1	−4.2	+3.1	−1.1	+0.5	−0.4	+9
72	+2.2	−1.6	+2.6	−4.3	+0.1	−0.2	+2
95	+1.4	−0.1	+1.0	−3.3	+0.7	−0.5	+1
117	+3.5	+0.7	−1.0	−2.6	−0.4	−1.0	−3
139	−0.3	+3.4	−5.5	−0.1	+0.9	−0.5	−6
162	−7.0	+9.4	−7.0	−7.1	+1.9	+1.7	−4

5 Results

5.1 Level-flight trim

Figure 2 and Table 4 compare the trims. From 50 to 162 kt the longitudinal cyclic is within 7.0 % of its travel, the collective within 7.0 %, the pitch attitude within 1.9° and the roll attitude within 1.7°. Main rotor power is within −6 to +9 %, the largest excess at 50 kt; the model reproduces the test's 930–950 hp minimum between 50 and 95 kt only from 72 kt up. The largest differences are at 162 kt: the collective 7.0 % low, the lateral cyclic 9.4 % right of the test and the pedal 7 % left, with the nose 1.9° high. The near-hover test points need 7–14 % more collective than the model at 22–32 kt and 3–4 % more than its hover at 1 kt; there the stabilator's schedule saturates at the model's 39° limit, where the test reads 44.5°.

5.2 Lateral-directional trim

At zero sideslip (Table 5) the model's lateral cyclic is within 0.30 in of the test and its bank within 1.3°. GenHel's downwash correction terms matter here: without them the model trimmed at up to 2.5° of left bank, a side force of some 900 lb missing near the CG at 100 kt, which neither the tail rotor nor the rotor's tilt can supply. Two effects are only partly reproduced: the bank swings by 2.3° between a 1500 fpm descent and a climb against 4.3° in the test, and moving the CG from BL −4.1 to +2.1 in costs −0.96 in of lateral cyclic against −1.48 in, so the model's roll moment per inch of cyclic is about 1.5 times too large, or the test's lateral CG shift smaller than its loading implies.

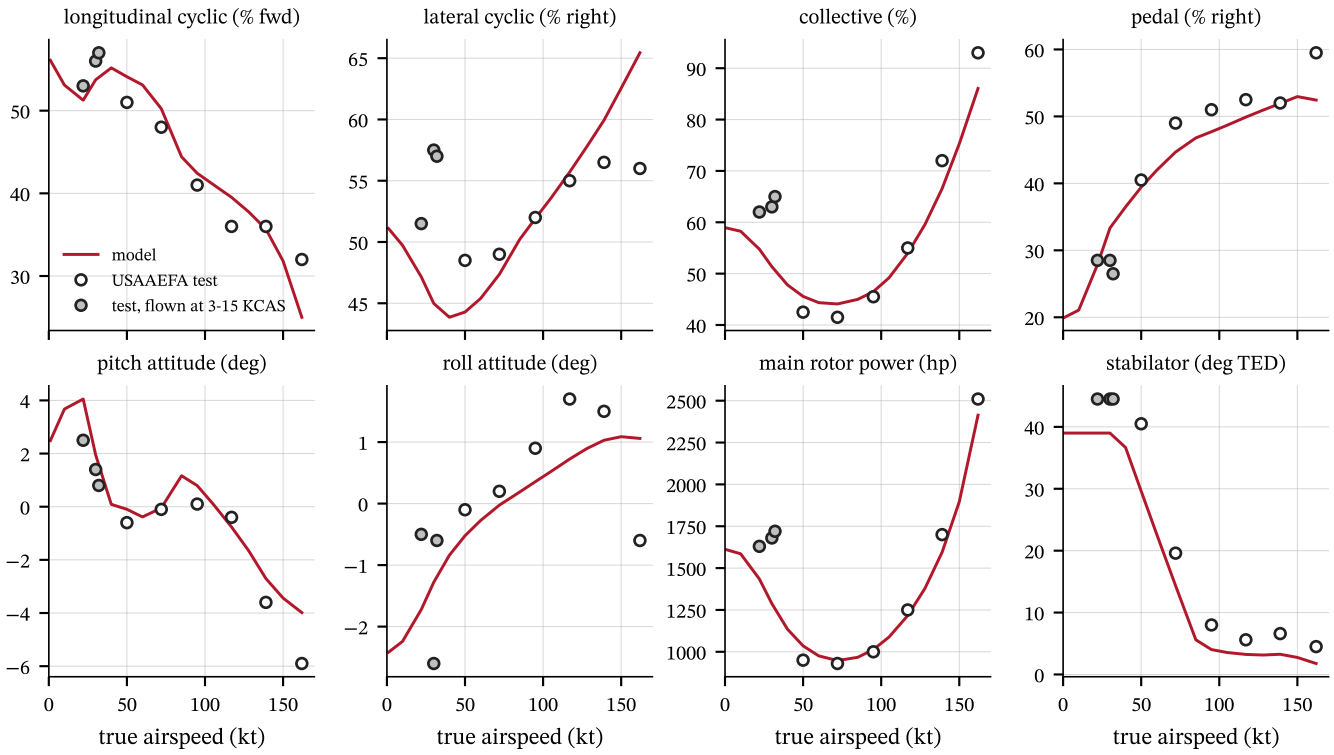


Figure 2: Level-flight trim against the USAAEFA test of TM-88360 fig. B1: 16 000 lb, FS 351, 5250 ft density altitude. Grey symbols: points plotted at 22–32 kt but flown at 3–15 KCAS. Controls are in per cent of travel; power is the main rotor's.

Table 5: Lateral-directional trim against USAAEFA 79-24 sweeps (zero sideslip, from the sweep through it) and ADA180404. Lateral cyclic in inches from full left, pedal in per cent of travel from full left.

	roll (deg)		lat. cyclic (in)		pedal (%)	
zero sideslip	test	model	test	model	test	model
60 kt level	0.1	−0.3	4.84	4.54	43	42
100 kt level	1.8	0.4	5.48	5.27	49	48
140 kt level	1.4	1.0	5.87	6.02	51	52
100 kt, 1500 fpm descent	3.2	1.5	–	4.93	49	50
100 kt, 1500 fpm climb	−1.1	−0.8	–	5.49	49	44
100 kt, CG BL +2.1 in	2.5	1.0	4.60	4.95	–	47
100 kt, CG BL −4.1 in	−0.7	−0.7	6.08	5.91	–	49

Wings level, sideslip (deg, + wind from the right): model 60 kt +1.6, 100 kt −1.1, 140 kt −1.4; 100 kt climb +2.0, descent −4.0. Test: −4 at 67 KCAS to −1 at 150 KCAS [5]; +5 in IRP climbs, −5 in 1000 fpm descents [8].

Table 6: OGE hover: model against the first-year aircraft's tethered-hover fit [7]; tail rotor against 77-17 fig. 10 (bracketed: outside its data).

weight	C_T	$C_P \times 10^5$		ratio	tail rotor
lb	$\times 10^4$	model	test		ratio
11 000	43.5	37.38	36.57	1.022	(0.90)
13 000	51.4	44.46	44.69	0.995	(0.88)
15 000	59.3	52.35	53.46	0.979	(0.88)
17 000	67.3	61.03	62.83	0.971	0.89
19 000	75.2	70.44	73.00	0.965	0.91
21 000	83.1	80.81	84.76	0.953	0.93
22 000	87.0	86.56	91.30	0.948	0.95

5.3 Power

GenHel applies the tip loss factor to the blade lift only. Its inflow is the momentum solution for the full disc, so the induced power it computes is ideal. The physical counterpart of the lost lift near the tip is a wake carried by the effective disc B^2A [17]: the induced velocity is $1/B$ times the full-disc value in hover and $1/B^2$ edgewise. The model applies this to both rotors by scaling the Peters–He forcing, and the tail rotor's momentum term, by $1/B^2$. The isolated rotor's figure of merit (Fig. 3d) then lies within 0.010 of a blade-element calculation with annulus momentum and Prandtl's distributed tip loss, and within 0.026 of the measured values; with the lift cut alone it is 0.028 higher.

The complete aircraft (Fig. 3a, Table 6) needs 0.965–0.995 of the measured engine power in the range of the test data and 0.948 at 22 000 lb; the deficit grows with thrust. The tail rotor needs 0.89–0.95 of its measured shaft power. In level flight (Fig. 3c) the model needs 2 to 10 % more power than the 77-17 data, 9 % at $\mu = 0.22$, while its main rotor power agrees with the 79-24 measurements of the same airframe (Table 4) within 3.6 % from 72 kt up once the latter's instrumentation drag (2.8 ft^2 [12]) is allowed for. The power budget (Table 7) shows nothing out of the ordinary: the airframe's drag area of

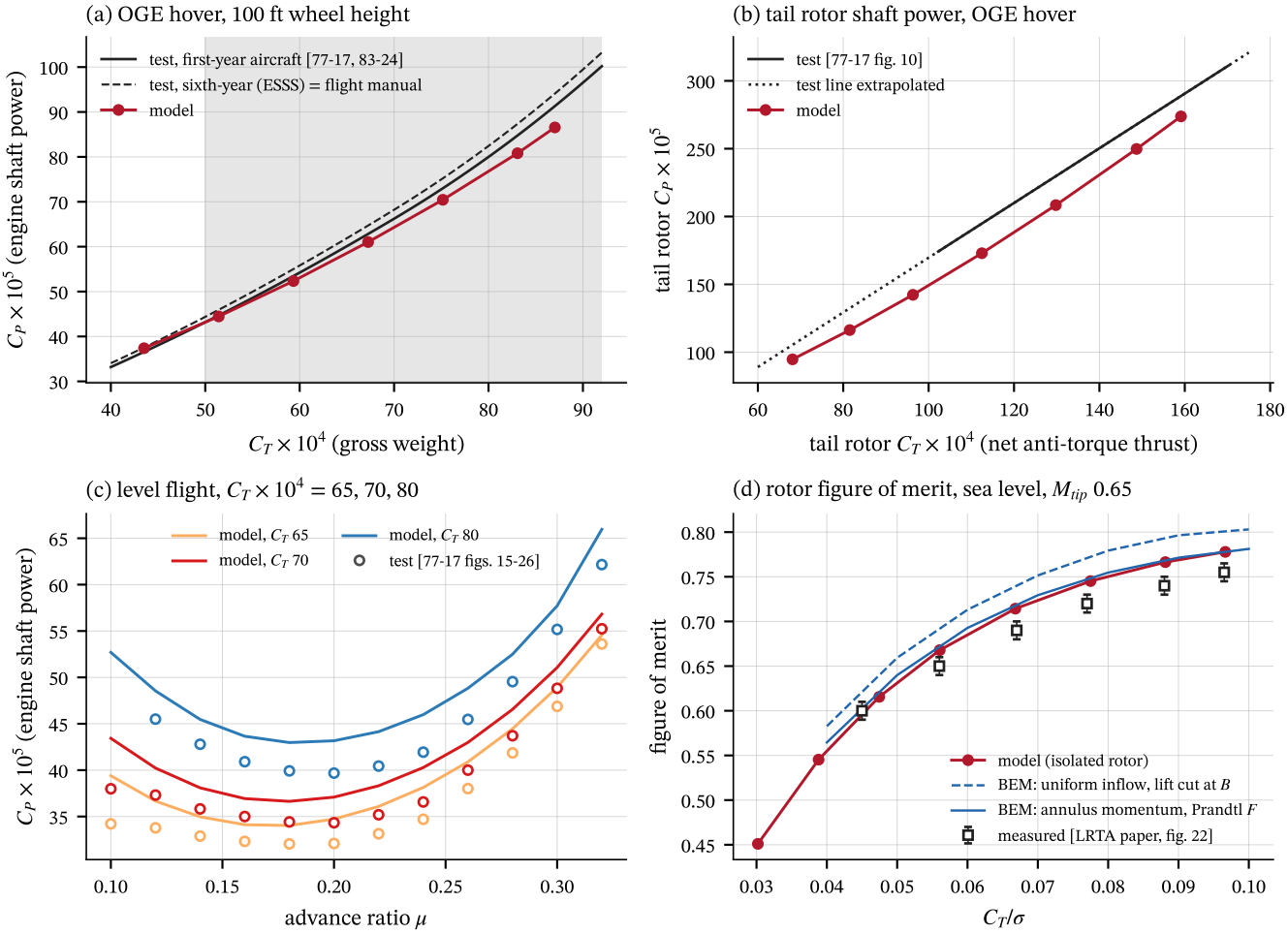


Figure 3: Power. (a) Out-of-ground-effect hover at the 2302 ft test site against the tethered-hover fits of 83-24 (shaded: test data range). (b) Tail rotor shaft power in the same hovers against 77-17 fig. 10. (c) Level flight against the faired data of 77-17 figs. 15–26 at 263 referred rpm. (d) Figure of merit of the isolated rotor, with two blade-element momentum (BEM) calculations of the same blade.

Table 7: Power budget: OGE hover at 17 000 lb (2302 ft site) and level flight at $C_T = 0.0070$ (5250 ft).

	hover	μ 0.22	μ 0.30
weight (lb)	17000	16886	16886
main rotor, induced (hp)	1362	408	308
blade profile drag	348	457	650
propulsive (airframe drag)	0	195	480
tail rotor	205	60	75
gearbox and accessories	117	98	109
engine shaft power (hp)	2032	1218	1622
induced power factor κ	1.06	1.13	1.13
airframe drag area (ft^2)	–	26.0	25.5
download (% of weight)	2.3	–	–

26.0 ft^2 matches Sikorsky's 25.7 ft^2 [12], induced power factors are 1.06 in hover and 1.13 in cruise, and the blades' profile drag is 43 % of main rotor power at $\mu = 0.22$, from section drag coefficients that agree with the SC1095 polars [9]. The pattern that remains, too little power in hover and too much in cruise against 77-17, survives any choice of the drag increment; it is the largest unexplained item of the model. The hover download is 2.3 % of weight, for which no UH-60 measurement was found.

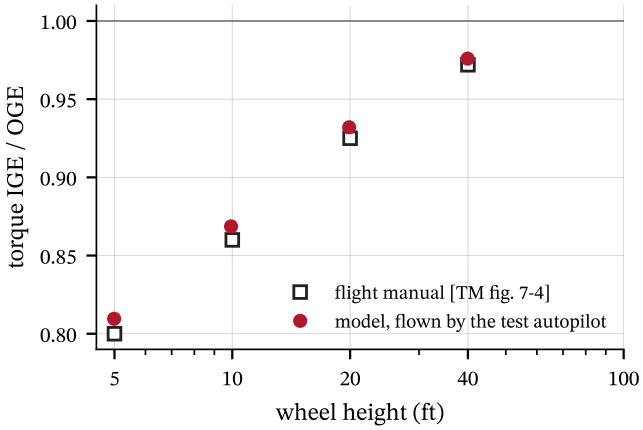


Figure 4: Hover torque in ground effect relative to 100 ft, model (flown by the test autopilot, 16 000 lb, sea level) and the operator's manual hover chart.

5.4 Ground effect

With the test autopilot holding the height (Fig. 4, Table 8), the torque ratios are within 0.010 of the operator's manual at every height; flights in FlightGear with the same autopilot recorded 0.81, 0.87, 0.93 and 0.98.

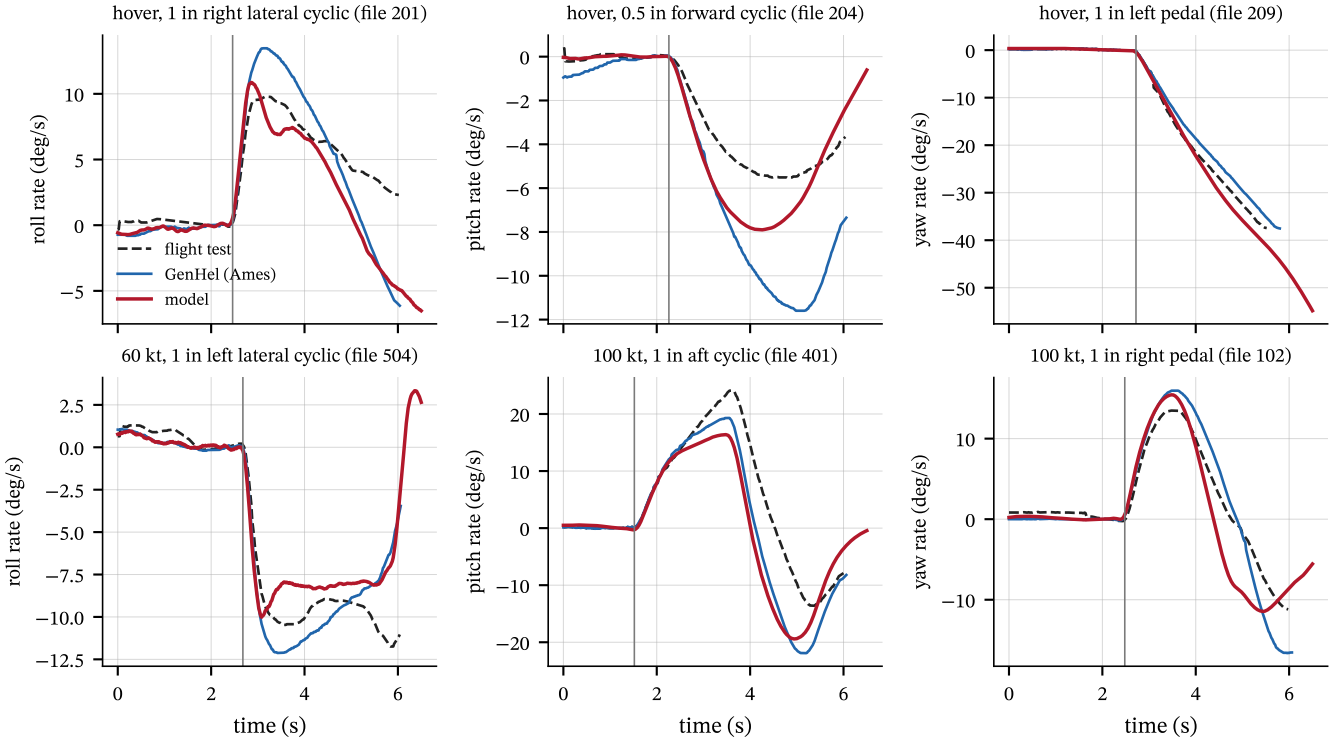


Figure 5: On-axis rate after control steps with the SAS off (TM-88360 app. D): flight test, GenHel as run at NASA Ames, and the model, as increments from trim. The vertical line marks the input.

Table 8: Ground effect: hover torque relative to 100 ft wheel height.

wheel height (ft)	model	flight manual	difference
5.0	0.810	0.800	+0.010
9.9	0.869	0.860	+0.009
19.9	0.932	0.925	+0.007
39.9	0.976	0.972	+0.004

5.5 Control response

Figure 5 shows six of the 17 cases and Table 9 all of them. The model's rms error is below GenHel's in 12 of 16 cases, with a mean of 2.14 deg/s against GenHel's 2.77 deg/s. In hover the model sits between GenHel and the test: GenHel's peak roll and pitch rates after cyclic steps are 38–106 % larger than the test's, the model's 11–43 %. The model's yaw rate after the pedal step in hover is 9 % larger than the test's. The clearest deficiency is the aft cyclic step at 100 kt: the test reaches 24.1 deg/s of pitch rate, GenHel 19.3 deg/s, the model 16.4 deg/s. At 100 kt both simulations also roll the wrong way after a pedal step, which neither reproduces.

5.6 Stability derivatives

The derivatives (Table 10) explain the weak pitch-up. The model's angle-of-attack stability is close to neutral, as the aircraft's (79-24 found the static stability negative to neutral), but its M_w of 0.0028 is a third of GenHel's 0.0092: the stabilator's restoring moment nearly cancels the destabilising contributions of rotor and fuselage. Z_w is −0.68 against −0.79, and a Pitt–Peters wake gives the same, so it is not the inflow model. Z_q has the opposite sign to GenHel's (−1.1 against 5.0): in hinged-rotor theory pitch rate tilts the disc but does not change its thrust in hub-plane axes, which the model follows. With the SAS on, 79-24 found a long-term aperiodic pitch divergence at 100 kt that the model does not show; it is stable with the pitch bias actuator on and close to neutral with it off.

6 Corrections found during validation

The model is the result of several rounds of validation; the changes that the comparisons forced are listed here because each removed a physical inconsistency rather than tuning a coefficient.

1. *Rotor wake on the airframe in hover.* GenHel puts the in-plane part of the wake's velocity at the fuselage, fin and tail rotor along the body's x axis. The first version used the direction of the in-plane flow instead, which is undefined in hover: a drift of 0.05 ft/s swung 6 ft/s of wash sideways onto the fin and tail rotor and kicked the aircraft in yaw. It now fades to the body axis below 5 ft/s. The hands-off hover with the SAS on became steady (heading within 0.7° in 30 s, against 93° before).
2. *Tip loss in the momentum balance* (Section 5.3): hover power was 4–9 % low and the figure of merit 0.03 high.
3. *Accessory power*: a placeholder of 30 hp was replaced by Sikorsky's 65.8 hp, and an unsourced section drag increment of 0.002 by the 0.0012 of a tripped boundary layer.
4. *Downwash correction terms*: GenHel's empirical wake-swirl forces on the empennage, defined in its motion module rather than the empennage module, were missed at first; without them the aircraft trimmed with 1.4–2.5° of left bank at zero sideslip.
5. *Test conditions*: the control responses were first replayed at the CG of fig. B1. At the CG of table C1, FS 359.5 for the 100 kt cases, the aft-step pitch rate rose from 11.4 to 17.5 deg/s.
6. *Ground effect*: an earlier version scaled thrust; the effect

Table 9: Control responses (TM-88360 app. D): peak on-axis rate in the 2.5 s after the input, and the rms error of the rate against the test over that interval (bold: model below GenHel). †The test pitch-rate trace of 509 could not be digitised reliably; excluded from the statistics.

				peak (deg/s)			rms error	
file	condition	input		test	GenHel	model	GenHel	model
201	hover	1 in right cyclic	p	9.8	13.5	10.9	2.4	**1.8**
202	hover	1 in left cyclic	p	−13.1	−20.7	−16.7	6.5	**2.0**
204	hover	0.5 in forward cyclic	q	−5.5	−11.4	−7.9	3.5	**2.1**
212	hover	1 in up collective	r	13.8	9.0	12.3	2.0	**1.3**
209	hover	1 in left pedal	r	−35.0	−31.9	−38.1	2.5	**2.0**
504	60 kt	1 in left cyclic	p	−10.5	−12.1	−10.0	1.3	1.5
509	60 kt	1 in aft cyclic	q	−26.8†	14.5	−15.8	13.2	16.5
405	60 kt	0.5 in down collective	r	0.9	1.0	−1.0	0.5	1.2
510	60 kt	0.5 in up collective	r	3.1	−1.3	−0.4	2.4	**2.2**
512	60 kt	1 in down collective	r	−2.3	−3.0	−1.8	0.9	**0.5**
502	60 kt	0.5 in right pedal	r	9.0	12.4	10.8	4.1	**1.4**
503	60 kt	1 in left pedal	r	−11.9	−13.2	−12.9	1.7	**1.2**
401	100 kt	1 in aft cyclic	q	24.1	19.3	16.4	3.9	5.9
706	100 kt	1 in forward cyclic	q	−7.6	−9.8	−10.5	3.1	**3.0**
404	100 kt	0.5 in up collective	r	1.2	−1.4	0.4	1.5	**0.8**
102	100 kt	1 in right pedal	r	13.5	16.0	15.4	2.4	3.9
103	100 kt	1 in left pedal	r	−12.2	−18.5	−16.8	5.5	**3.5**

Table 10: Longitudinal derivatives at 100 kt, 16 400 lb, FS 360.4 (TM-85890): GenHel, and the model with the pitch bias actuator and stabilator active, and with them held (rotor and airframe only). Per unit mass or pitch inertia; δ_e inch aft, δ_c inch up.

		GenHel	model	
	unit	[2]	PBA, stab.	aero only
X_u	1/s	−0.0324	−0.0412	−0.0390
X_w	1/s	0.0643	0.0494	0.0492
X_q	ft/s/rad	2.788	3.673	2.802
Z_u	1/s	−0.0089	0.0303	0.0462
Z_w	1/s	−0.790	−0.678	−0.680
Z_q	ft/s/rad	4.981	0.667	−1.062
M_u	rad/s/ft	0.0025	0.0023	−0.0027
M_w	rad/s/ft	0.0092	0.0037	0.0028
M_q	1/s	−1.606	−2.827	−1.757
X_{δ_e}	ft/s^2/in	−1.083	−1.259	−1.258
Z_{δ_e}	ft/s^2/in	−6.138	−4.794	−4.852
M_{δ_e}	rad/s^2/in	0.459	0.555	0.542
X_{δ_c}	ft/s^2/in	0.646	0.696	0.705
Z_{δ_c}	ft/s^2/in	−9.630	−8.177	−7.918
M_{δ_c}	rad/s^2/in	0.0951	0.0766	0.1683

is on the induced velocity, fitted to the hover chart and carried into forward flight with the measured interference ratios.

7. *Ground contact*: wheels alone let the fuselage sink through the ground when the aircraft rolled over, and JSBSim's wheel force diverges as a strut lies along the ground. Structure contacts from the 3-D model and a rule that takes the wheels out of play beyond 60° of tilt fixed both.

7 Assessment

Table 11 summarises. The model reproduces the Black Hawk's trim, its hover and ground-effect power and its short-term responses at least as well as GenHel, from which it was

Table 11: Summary: ✓ within the test's scatter, ~ partly, × not reproduced.

Trim controls and attitudes, 50–139 kt	✓
Trim at 162 kt (collective, lateral cyclic, pedal)	~
Lateral-directional trim at zero sideslip	✓
Bank change between climb and descent	~
Hover power, OGE and in ground effect	✓
Rotor figure of merit	✓
Tail rotor power in hover	~
Cruise power (one test ✓, the other ×)	~
Responses to cyclic, collective, pedal in hover and at 60 kt	✓
Pitch-up after an aft step at 100 kt	~
Roll after a pedal step at 100 kt	×
Long-term pitch divergence with SAS (79-24)	×

derived, and in hover better. Its open items share one direction: in pitch at speed the model is too stable. A third of GenHel's M_w, too small a pitch-up after an aft step and no long-term divergence point to the same missing destabilising moment, and Section 5.6 locates it in the balance of rotor, fuselage and stabilator rather than in any single term. The other open items are the cruise power against one of the two tests, the tail rotor's hover power (the test lumps the fin's interference into its net thrust), and the lateral moment per inch of cyclic. Retreating-blade stall limits the model at high thrust and speed ($C_T = 0.009$ above $\mu = 0.28$); blade torsion and dynamic stall are not modelled.

8 Conclusions

- A blade-element model of the UH-60A with individual blades, a finite-state dynamic wake, GenHel's airframe and control laws, and engines runs in plain JSBSim, generated from data. It is the flight model of the FlightGear UH-60A.
- Against the US Army flight tests it reproduces level-flight longitudinal cyclic and collective within 7.0 % of travel up

to 162 kt, hover power at 0.965–0.995 of the measured over the test's range, ground effect within 0.010 and the SAS-off control responses with a smaller error than GenHel.

- Applying tip loss to the momentum balance as well as the lift brings the rotor's figure of merit and the aircraft's hover power into line with the measurements; GenHel's lift-only tip loss leaves the induced power ideal.
- The pitch stability at speed is too strong and the cruise power against the 77-17 data too high; both are documented, not tuned away.

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